

AI Provider Cost Analysis

Comparing Anthropic Claude, OpenAI GPT, and Google Gemini for Hype's AI features — March 2026 pricing

1. Model Pricing Comparison

Only showing models relevant to Hype’s use case (short prompts, structured output, creative text). Excluding reasoning-heavy models (o3, o1) and flagship models overkill for this task.

MODEL	PROVIDER	INPUT / 1M TOKENS	OUTPUT / 1M TOKENS	CONTEXT	NOTES
GPT-4.1 nano	OpenAI	\$0.10	\$0.40	200K	Cheapest usable model
Gemini 2.5 Flash-Lite	Google	\$0.10	\$0.40	1M	FREE TIER 1,000 RPD
GPT-5 nano	OpenAI	\$0.05	\$0.40	—	Newest, cheapest
Gemini 2.5 Flash	Google	\$0.30	\$2.50	1M	FREE TIER 250 RPD
GPT-4.1 mini	OpenAI	\$0.40	\$1.60	1M	Strong instruction following
Claude Haiku 4.5	Anthropic	\$1.00	\$5.00	200K	Best personality/voice quality
Gemini 2.5 Pro	Google	\$1.25	\$10.00	1M	FREE TIER 100 RPD
GPT-4.1	OpenAI	\$2.00	\$8.00	1M	Full flagship quality
Claude Sonnet 4.6	Anthropic	\$3.00	\$15.00	1M	Premium creative quality

2. Hype AI Features — Token Budgets

Tonight AI Planner SHOWSTOPPER

User picks mood + group size + stops. System prompt includes 20–50 venue summaries from Supabase. Claude returns a structured evening itinerary with 3–5 stops, descriptions, and timing.

Input: ~2,500 tokens (system prompt + venue context + user preferences)
Output: ~1,200 tokens (structured plan with descriptions)
Calls per use: 1–2 (generate + optional regenerate)

AI City Pulse HIGH IMPACT

Home hero shows a dynamic, AI-written summary of what's happening in Sarajevo right now. Based on today's events, weather, time of day. Refreshes every few hours or on pull-to-refresh.

Input: ~1,500 tokens (today's events + weather + time context)
Output: ~200 tokens (2–3 sentence blurb)
Calls per use: 1 (cached for several hours)

Natural Language Search

"Good coffee near Baščaršija" → structured search filters. Interprets intent, maps to categories/neighborhoods/moods, returns Supabase query parameters.

Venue/Event Descriptions

Enrich thin database descriptions with engaging, tourism-ready copy. Batch job, not real-time. Run offline, store results.

Input: ~800 tokens (system prompt + user query + available filters)
Output: ~150 tokens (structured JSON filters)
Calls per use: 1 per search

Input: ~500 tokens (venue data + style prompt)
Output: ~300 tokens (description)
Calls per use: Batch — one-time per entity

3. Cost Per User Session

Assumes a typical session: 1 planner call, 1 city pulse (shared/cached), 2 NL searches. Total: ~4 calls, ~7,300 input tokens, ~2,750 output tokens.

MODEL	PER SESSION	1K SESSIONS	10K SESSIONS	100K SESSIONS
GPT-5 nano OpenAI	\$0.0015	\$1.46	\$14.65	\$146.50
GPT-4.1 nano OpenAI	\$0.0018	\$1.83	\$18.30	\$183.00
Gemini 2.5 Flash-Lite Google	\$0.0018	\$1.83	\$18.30	\$183.00
GPT-4.1 mini OpenAI REC	\$0.0073	\$7.32	\$73.20	\$732.00
Gemini 2.5 Flash Google	\$0.0091	\$9.07	\$90.68	\$906.75
Claude Haiku 4.5 Anthropic	\$0.0211	\$21.05	\$210.50	\$2,105.00
Gemini 2.5 Pro Google	\$0.0366	\$36.63	\$366.25	\$3,662.50
Claude Sonnet 4.6 Anthropic	\$0.0631	\$63.12	\$631.25	\$6,312.50

4. Monthly Cost by User Scale

Assumes 1 session/user/day on average. City pulse cached (shared across users, ~8 calls/day total).

PILOT

100 users

~3,000 sessions/month

GPT-5 nano	\$4.40
GPT-4.1 mini	\$21.96
Claude Haiku 4.5	\$63.15
Gemini 2.5 Flash	\$27.20
Gemini Free Tier	\$0.00 *

GROWTH TARGET

1,000 users

~30,000 sessions/month

GPT-5 nano	\$43.95
GPT-4.1 mini	\$219.60
Claude Haiku 4.5	\$631.50
Gemini 2.5 Flash	\$272.03
Best value	GPT-5 nano: \$44/mo

SCALE

10,000 users

~300,000 sessions/month

GPT-5 nano	\$439.50
GPT-4.1 mini	\$2,196.00
Claude Haiku 4.5	\$6,315.00
Gemini 2.5 Flash	\$2,720.25
Best value	GPT-5 nano: \$440/mo

* Gemini free tier can handle ~200 sessions/day — covers the full pilot.

Recommended Strategy: Multi-Provider Mix

Don't lock into one provider. Use the **best model for each feature** based on quality needs and cost:

FEATURE	RECOMMENDED MODEL	WHY
Tonight AI Planner	GPT-4.1 mini	Best instruction-following for structured itineraries. Strong creative output. \$0.007/plan.
City Pulse (Home hero)	Gemini 2.5 Flash-Lite	Cached result shared across all users. Cheapest option works fine. Near-free at scale.
NL Search	GPT-4.1 nano	Structured JSON output from short queries. Fastest, cheapest option that's reliable.
Batch Descriptions	Claude Haiku 4.5	Best prose voice for tourism copy. One-time batch cost, not per-session. Use batch API for 50% off.

Blended session cost with this mix: ~\$0.004/session (\$4/1K sessions, \$120/month at 1K users)

For the presentation demo: Use Gemini 2.5 Flash free tier for everything — zero cost, 250 requests/day is more than enough for a demo. Swap to the optimized mix when real users arrive.

Key insight for the tourism board: AI features cost pennies per user. At 1,000 monthly active users, the entire AI layer costs less than a single coffee in Sarajevo per day. This is not a cost problem — it's a capability unlock. The board should focus on what it enables (personalized discovery, intelligent recommendations, dynamic content) not what it costs.